



331SC

Cut Operation Manual

Graphic Whizard
V1 October 29, 2021
Mike@graphicWhizard.com

INDEX

Chapter 1 INTRODUCTION	1
1.1 Preface	1
1.2 Specification	1
Chapter 2 SAFETY	2
2.1 Environment	3
2.2 Do's and Don'ts	3
2.3 Cautions	4
Chapter 3 MAIN PARTS & ASSIES	5
Chapter 4 INSTALLATION	7
4.1 Parts Installation	7
4.1.1 Outfeed Part	7
4.1.2 Extension table	7
4.1.3 Place the machine	7
4.1.4 Power Socket and Switch	7
Chapter 5 OPERATION	9
5.1 Key Panel Introduction	9
5.2 Screen Introduction	10
5.2.1 Welcome Screen	10
5.2.2 Ready Screen	10
5.2.3 Recall Job Screen	10
5.2.4 Factory Template	10
5.2.5 Set Paper Length	10
5.2.6 Sniper Cut	11
5.2.7 Sniper Fine	11
5.2.8 Smart Input	11
5.2.9 Input Cut Data	11
5.2.10 Set Shrink/Stretch Ratio	11
5.2.11 Horizontal Shift Screen	12
5.2.12 Card Length Offset Screen	12
5.2.13 Gutter Offset Screen	13
5.2.14 Save Job Screen	13
5.2.15 Speed Screen	13
5.3 Smart Input Mode	14
5.3.1 Introduction of the sheet data	14
5.3.2 Input Lead Trim	14

5.3.3 Input Card Length & Gutter	14
5.3.4 Tail Trim Error	15
5.4 Register mark	15
5.4.1 Register Mark specification	15
5.4.2 How to use a Register Mark?	15
5.5 Sniper	16
5.6 Hardware Setting	17
5.6.1 Side Guide Setting	17
5.6.2 Skew Adjustment	18
5.6.3 Sitters Setting	18
5.6.4 Central Feeding setting	19
5.7 Only Trim Sheet Sides	20
5.7.1 Place Slitters	20
5.7.2 Set Cut Data	20
Chapter 6 Factory Template	21
6.1 Metric Version	21
6.2 Imperial Version	23
Chapter 7 Trouble Shooting	24
7.1 C-3 Paper Jam at Outfeed Roller	24
7.2 E-2 Cut Motor Error	25
7.3 Card Cutting non-precise	25

Chapter 1INTRODUCTION

1.1 Preface

331SC sniper Swift Cut is a enter level machine with cost effecton that can cut big sheet into small card at flexible size.

The cutter position can be programmed and slit position is manual adjusted.

The sniper function is embeded to ensure the precision.

The machine is perfect solution name card, coupon, poster card, flyer

This manual only applies to 331SC Sniper-Swift Cut machine.

Before using the machine, please read the instruction manual carefully first. Please keep the manual properly so that consulting it at any time in the future. Any question while operating please contact our technicians.

1.2 Specification

Item	Parameter	
Package Dimensions	Length*Width*Height (mm)	740*610*460
Machine Dimensions	Length*Width*Height (mm)	973x572x322
Net Weight		44kg
Package Weight		50 kg
Power Input		AC220V,50/60Hz, 150W
Output Size	Max (W*L) (mm)	318×640
	Min (W*L) (mm)	55×48
Infeed Size	Max (W*L) (mm)	330×650
	Min (W*L) (mm)	297×210
	Thickness	150g~350g/0.15~0.35mm
	Paper Waviness	±2mm
Cutter	Max Qty for Cutter per Sheet	32
	Accuracy for Cutting	±0.3mm(except for special paper)
Slit	Slitters	4
	Side Trim	3.2~38mm
	Gutter Slit	8mm
	Accuracy for Slitting	±0.3mm(except for special paper)
Feeding	Middle Infeed	Hand Feed
User Template Storage		30
Register Mark		Y Direction only
Sniper technology		See page

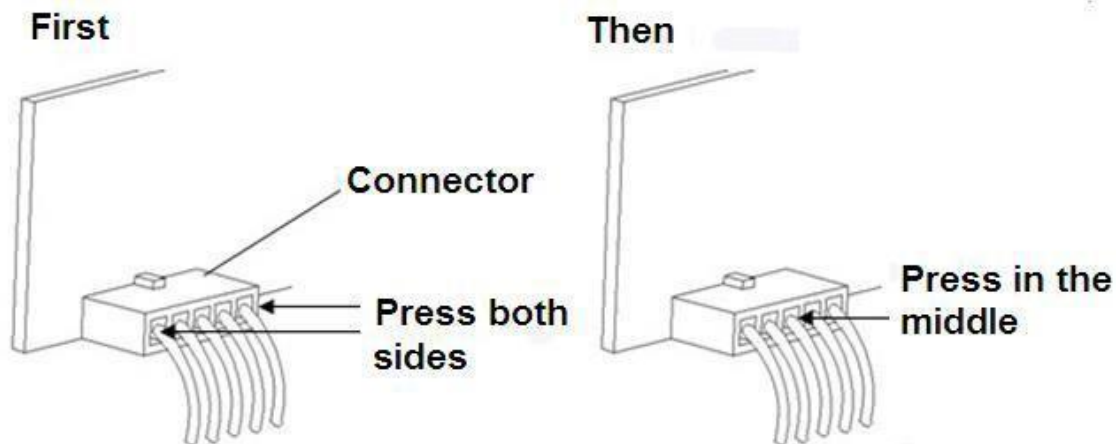
Note: The machine is keep upgrading, specification and information in this manual are updated as per the change without notice.

Chapter 2 SAFETY

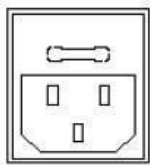
Please shut down the machine before the maintenance.

In order avoid any mistake when re-connecting wires back, we advise you to mark each connector before replace of any electrical component.

Mind that every connection should be firm, first hold both sides of the connector, and then, in the middle, not firm connection is possible to damage the electrical components.

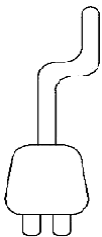


Power supply



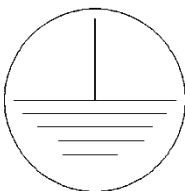
AC input, Voltage/Frequency is: AC220V/50-60Hz.

! If the voltage or frequency are not in the range, the machine may not be used normally or even may



Power cord, Over 10A current is endurable

! 1. Do not remove the cord when the machine is running, illegal operation may cause component damage or machinery jam



Ground symbol. If you see this symbol in any place, please do not touch it anyhow, or uninstall it.

! If the ground is not well connected, it may cause human body injury.

2.1 Environment

Temperature: 10°C to 35°C

Humidity: 30% to 70%

Altitude: Below elevation 1000m

There is no corrosiveness gas, flammable gas, oil mist and so on in room

2.2 Do's and Don'ts



Do-Read this manual and fully understand before the operation.



Do-Check the *plug and machine voltage and frequency* to your main supply, and that the socket has a correct working earth lead for this single insulated machine



Douse at least a 10 amp power source at 220 V. (20Amp/100V).



Do make sure all safety covers are in place. The top covers have an interlock switch which will disable the unit if removed



Do open or close the cover slowly



Do contact the local maintenance center before you are about to move the machine.



Do disconnect the power before clean the inner side.



Do unplug the cord if you won't use the machine for a long while,



Do be careful of the blade edge



Don't install the machine on an unstable ground.



Don't operating with wet hand, especially plug or unplug the cord.



Don't wear long hair, loose fitting clothes or put your fingers into the creasing unit nip, while the operation..



Don't place any receptacles with any liquid on any surface of machine.



Don't put other pieces, especially tiny pieces on loading table.



Don't alter or uninstall the machine, unless by *Dumor* authorized engineer.



Don't touch any running parts while running



Don't shut down the machine while running.



Don't put heavy matter on machine or shock it.



Don't put the side guide under the feeding system.

2.3 Cautions

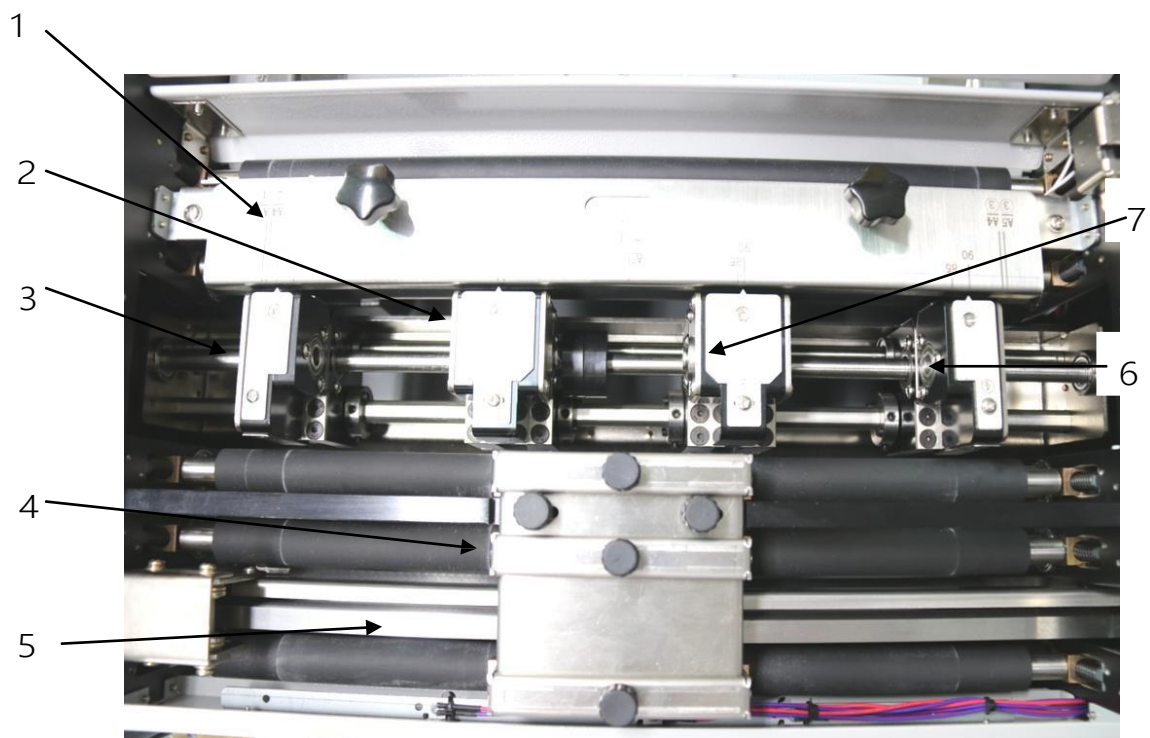
Be careful of any metal or flammable thing in internal machine, or it may cause fire or electronic shock. If it happens, first shut down the power, disconnect the cord, and then contact the technician.

If machine becomes heat, smoke, or smelly, shut down at once, disconnecting the cord, and contact the maintenance staff.

Chapter 3 MAIN PARTS & ASSIES



No.	Description	Remark
1	Card Collector	Collect finished business card
2	Outfeed Tray	Collecting finished Bigger Cards or Sheets
3	Waste ejection	Where paper strip come out
4	Control panel	Screen and keyboard
5	Side guide	To fix paper while feeding
6	Safety cover	To protect inner machine



No.	Description	Remark
1	Position indicator Plate	Fixing all the slitters distance. (Standard is 85mm and 90mm or 3.5inch and 5.5 inch)
2	Slitter #2	Slit with fixed gutter, gutter is 8 mm or 3/8 inch
3	Slitter #1	Slit the side edge of paper
4	Lag wheel Set	Adjust the roller pressure
5	Cross Cutter	Cut paper in cross direction
6	Slitter #4	Slit the other side edge of paper
7	Slitter #3	lit with fixed gutter, gutter is 8 mm or 3/8 inch



No.	Description	Remark
1	Side guide(L)	
2	Extension table	
3	Ruler	Reference of best position for different paper size
4	Side guide(R)	
5	Set screw	Loosen to enable movement of side guides

Chapter 4 INSTALLATION

4.1 Parts Installation

4.1.1 Outfeed Part

Hang the outfeed tray onto machine and place the business card collector on the proper position. As well as the deflector

Business card

Collector

tray



4.1.2 Extension table

Fix the extension table as per following picture :



4.1.3 Place the machine

Place the machine on some table and place a stripbox in front of it. So that the strips will drop off the box.

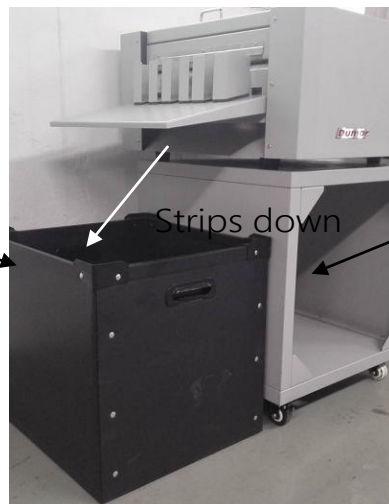
Stripbox

(End user

prepare ,

Not

standard



Strips down

Stand
(Option)

4.1.4 Power Socket and Switch

AC input, Voltage/Frequency is: AC220V/50-60Hz.



Switch (O is Off , I is

Socket Fuse: 3.15A

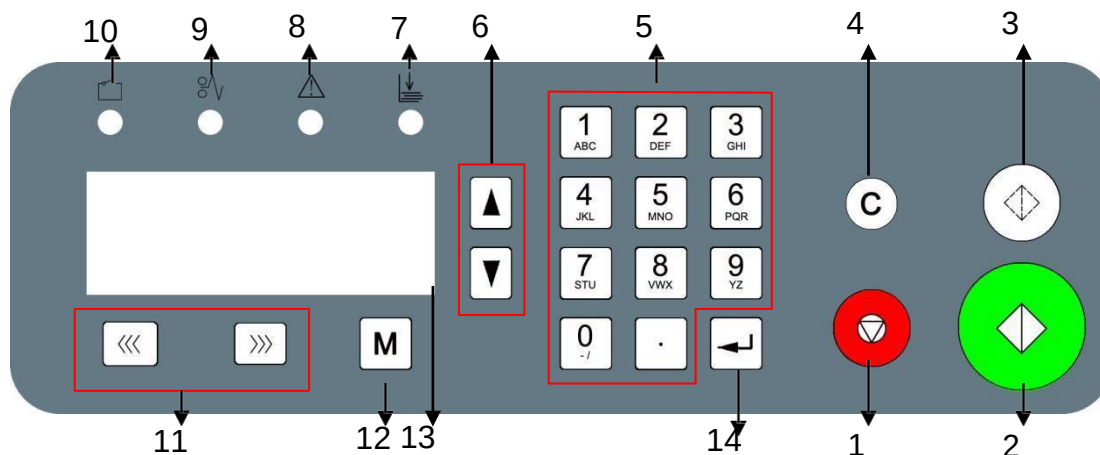


Cautions: Don't operate machine with wet hand.

Chapter 5 OPERATION

5.1 Key Panel Introduction

The control panel consists of keys and screen, as per following picture.



No	Key	Description
1	STOP	Stop the machine.
2	START	Run the machine.
3	TEST	Machine will stop after one sheet
4	CLEAR	Clear the data or clear the error display
5	NUMBERS	Input data
6	SCROLL	Turn the page
7	INDICATOR- NO PAPER	"C-4 No paper or Infeed Jam", but it's unavailable for this machine. No C-4 warning.
8	INDICATOR- MOTOR ERROR	If the crease motor locks, it lightens and returns an error report "E-1 CR Motor Error"
9	INDICATOR- PAPER JAMMED	If the paper is jammed inside the machine, it lights and returns an error report. "C-2 Jam at infeed" or "C-3 Jam at outfeed"
10	INDICATOR- SAFETY COVER	If the safety cover is not well setting, it lightens and returns an error report. "C-1 CR cover open"
11	FORWARD/REVERSE INCREASE/DECREASE	When paper jams, press them to move rollers, so that the paper can be easily cleared. Or increase/decrease the parameters for image compensation
12	MODE	Change the modes
13	SCREEN	Display information.
14	ENTER	After the data input, press it to confirm.

5.2 Screen Introduction

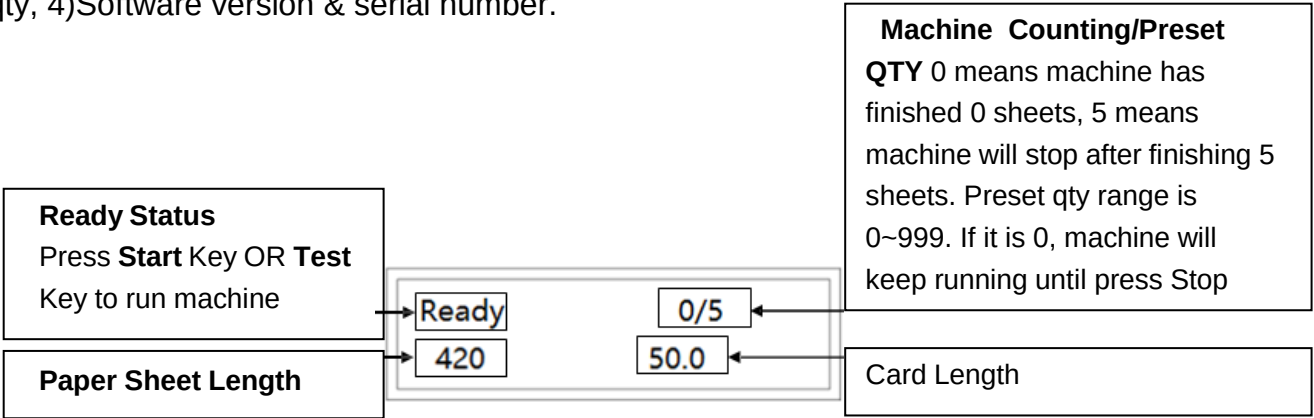
5.2.1 Welcome Screen

Turn on machine, the screen will show a welcome message, then enter Ready Screen.



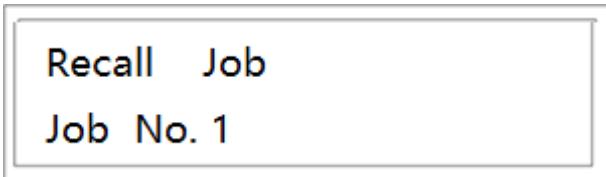
5.2.2 Ready Screen

Scan all information of the Ready Screen by pressing Page Down Key . The information mainly is divided into 4 kinds except the main ready screen: 1)Cut data, 2)Total sheets, 3)Cut qty, 4)Software version & serial number.



5.2.3 Recall Job Screen

Recall the user saved template. Input 1~30 and press Enter to take effect. The Job recalled should be saved as per 5.2.14



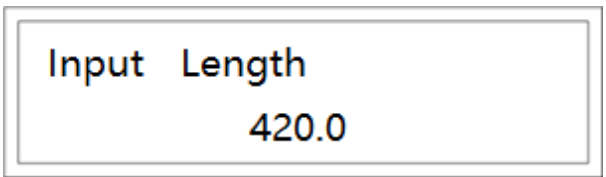
5.2.4 Factory Template

Press Mode Key  once to enter factory template screen. Press Page Up Key  or Page Down Key  to scan all factory templates. And then press Enter Key to select template and take effect. There are 16 templates for selection.



5.2.5 Set Paper Length

Input the paper sheet length: 210~650mm.



After length is set, Screen goes to setting Register mark.
If 1 on is chosen, it goes to another screen to set Mark distance. For more information please

refer to 5.1, then goes to Smart Input 5.2.8
If 2 off is chosen, it goes to Smart Input 5.2.8

REG.MARK

1=ON* 2=Off

Set REG.MARK

Distance:5.0

5.2.6 Sniper Cut

Sniper Cut screen enable the cutting against the mark position read by sensor, in which way, the precision of cutting can be ensured by printing precision
For more details please refer to 5.4

Sniper Cut

1=On* 2=Off

If 1 on is chosen, it goes to sniper fine 5.2.7, then goes to Smart Input 5.2.8
If 2 off is chosen, it goes to Register mark as per 5.2.5, then goes to Smart Input 5.2.8

5.2.7 Sniper Fine

Sniper fine defines accuracy compensation when the cutting is constantly different from the mark we preset.

Sniper fine

+0.3mm

5.2.8 Smart Input

Smart input is a quick job creating function. And it is a step by step guiding setting for this , please refer to 5.4

5.2.9 Input Cut Data

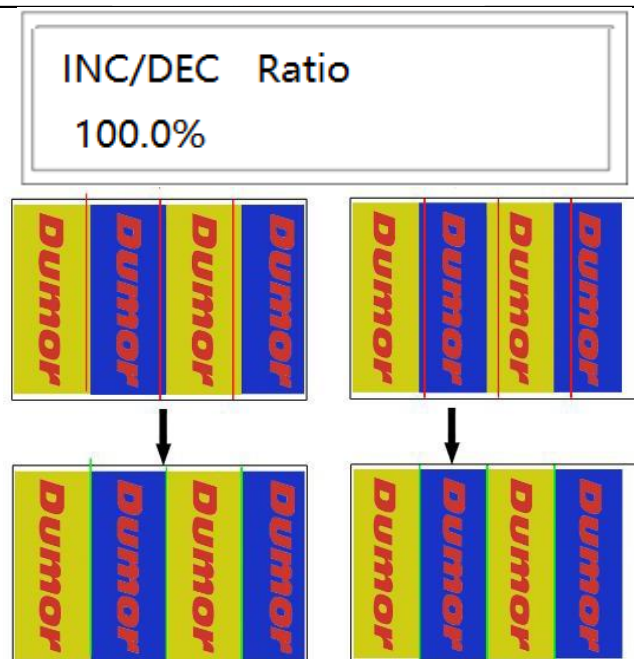
Input the cut data. Input value and press Enter Key  to take effect. Max qty is 32. Scan all the 32 cut data from 1 to 32 by pressing Page Down Key . There are 8 Screens for it.

Input Cut Data

1)10.0 2) 60.0

5.2.10 Set Shrink/Stretch Ratio

The function is aimed to settle image shrink/stretch caused by printing. Press Increase Key  to increase the ratio by 0.1% or press Decrease Key  to decrease the ratio by 0.1%. And press Enter to take effect

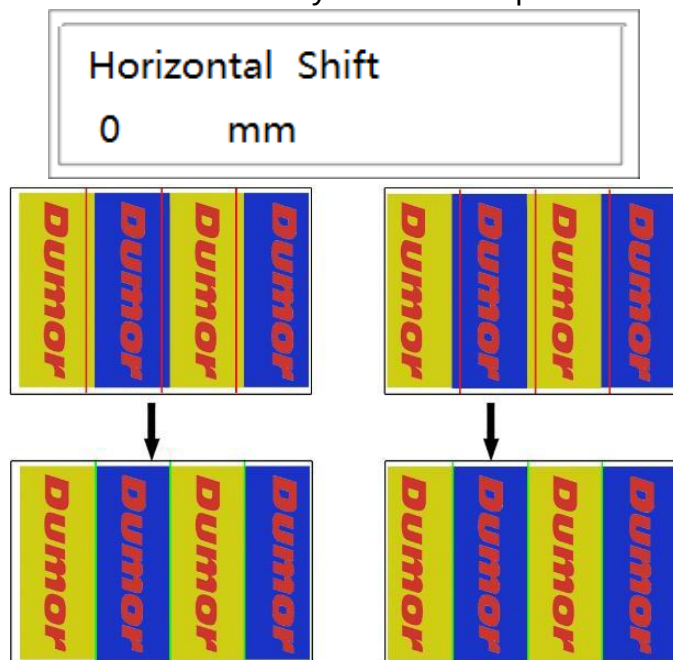


5.2.11 Horizontal Shift Screen

The function is aimed to settle image shift caused by printing.

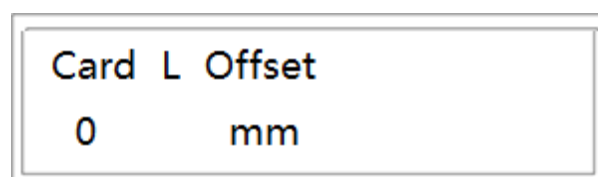
Press Increase Key  to increase the value by 0.1mm or press

Decrease Key  to decrease the value by 0.1mm. And press Enter to take effect



5.2.12 Card Length Offset Screen

The function is aimed to settle issue that all the true cards sizes are different from theory card size and the difference is the same. Press Decrease Key  to decrease the value by 0.1mm or press Increase Key  to increase the value by 0.1mm. And press Enter  to take effect



5.2.13 Gutter Offset Screen

Press Decrease Key  to decrease the value by 0.1mm or press Increase Key  to increase the value by 0.1mm. And press Enter to take effect.

Gutter	Offset
0	mm

Note: All the Image compensation functions 5.2.3.7~5.2.3.10 range is limited by First Cut (3.2~15mm), Gutter Cut (3~15mm) and Last Cut (5~15mm).

5.2.14 Save Job Screen

Save the user template by inputting value 1~30 as file name. The max storage is 30 templates. The Job No. can be recalled as per 5.2.3

Save	Job
Job	No.1

5.2.15 Speed Screen

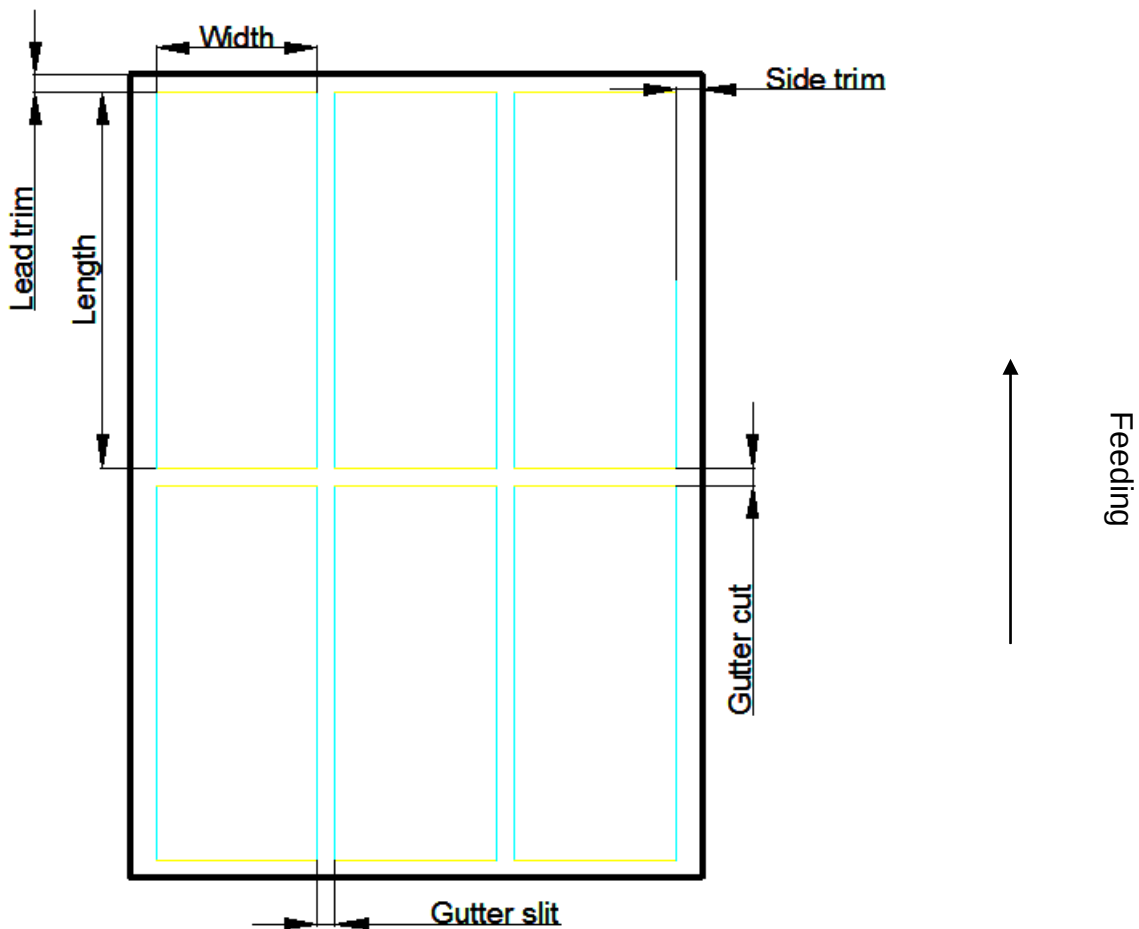
Select the speed mode of machine. 1 is low speed; 2 is high speed. And press Enter to take effect.

The default state is low. When restart machine, the speed is low.

Speed
1 = Low* 2 = High

5.3 Smart Input Mode

5.3.1 Introduction of the sheet data:



Example:

Card size: 90mmx200mm, Paper size: SAR3, thickness: 300gsm.

Lead trim: 30mm; Gutter cut: 10mm.

So the parameters for creating a job are: Lead Trim, Card Length, Gutter Cut

5.3.2 Input Lead Trim

Press Mode Key **M** for 6 times from Ready screen to enter Smart Input. Input value 30 and press Enter to confirm, The Trim range is 3.2~48. Since the first cut range is 3.2~15, the lead trim will be divided into some cuts which is in the range from 3.2 to 15. The lead trim 30 will be divided into two cuts 15 wide.

Smart Input	
Trim : 30.0	
1) 15.0	2) 30.0
3) 230.0	4) 240.0

If the Trim is over 15mm, more than 1 cut will be auto generated to make each cut less than 15mm

5.3.3 Input Card Length & Gutter

Input value 200 and press Enter to confirm. Input value 10 and press Enter to confirm. The Gutter Cut range is 3~15.

Card L : 200

Gutter : 10

5.3.4 Tail Trim Error

Since the last cut range is 5~15, if the tail (last cut) is out of the range, it will show Tail Trim Err. Press any key returning to Ready screen.

Tail Trim Err

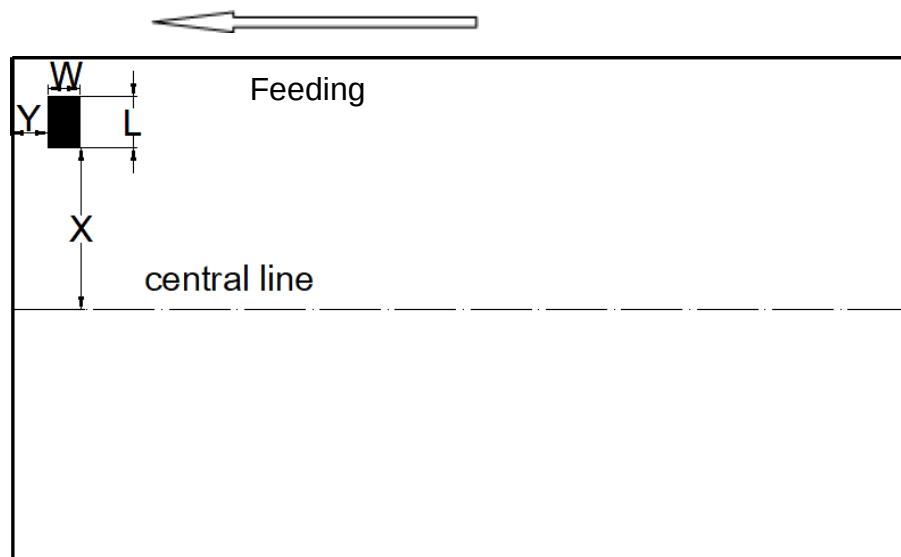
Any Key Return

5.4 Register mark

The register mark is much better for machine to position the images since that the register mark and images relative position is the same which could get rid of the shift caused by printing.

5.4.1 Register Mark specification

Create a mark as per following drawing



1. Register cut mark (Y) distance should be between 3~12 mm from leading edge. which is settable according to template.

Distance (X) should be 152mm

2. The black mark itself should be more than 3*12 ($L \geq 12\text{mm}$, $W \geq 3\text{mm}$)

3. All area above the mark should be blank, because other color may affect the mark reader.

4. The register mark reference point is the top edge of the mark.

Note: The central line position is according to the central position is found as per 5.6.4. if the infeed is not central, the mark may not work.

5.4.2 How to use a Register Mark?

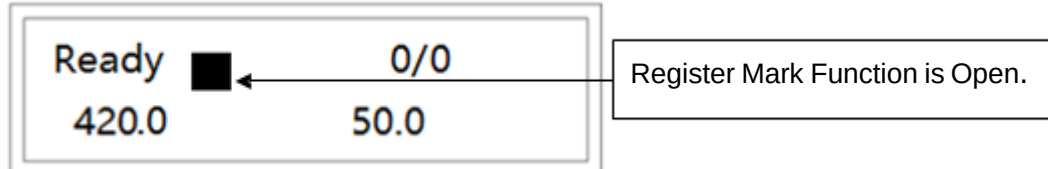
1. The Register screen will be shown after Paper length is set (5.2.5), or after Sniper function is not chosen (5.2.6). After Register mark is turned on, you have define the mark distance which is exactly value Y of above. Lets say it is 4, input 4 and press Enter to confirm. Then it will enter the next screen Smart Input Screen.(5.2.8)

When the print sheet input into the machine, it will read the real distance Y of mark, if machine find Y has any different of 4, the difference will be added into each cut thereafter to compensate the print shift.

Card L : 200

Gutter : 10

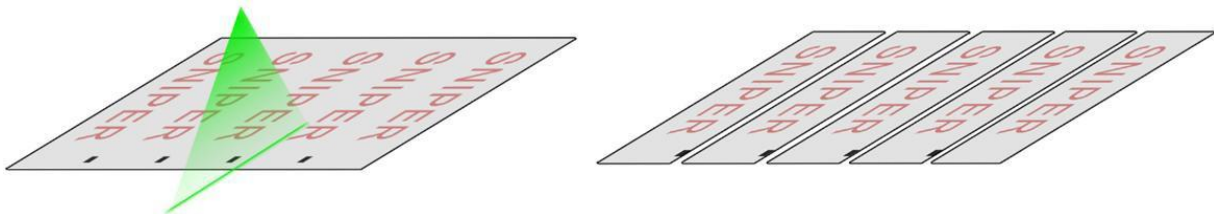
If no need to create a job, just press Stop to go back to Ready Screen.



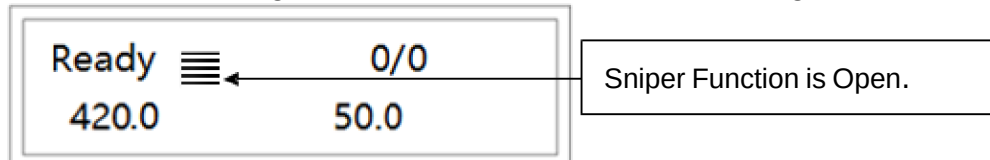
Note: The Register Mark is **only** used for user template rather than factory template.

5.5 Sniper

Sniper function request a set of mark print on paper so when it pass into the machine, the mark position will be read and cutting will be relatively made on the fly, the precision will be only decide by the print, not by the mechanical

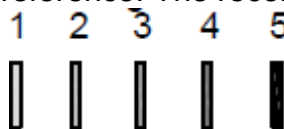


When Sniper function is working, the screen mark will be as following

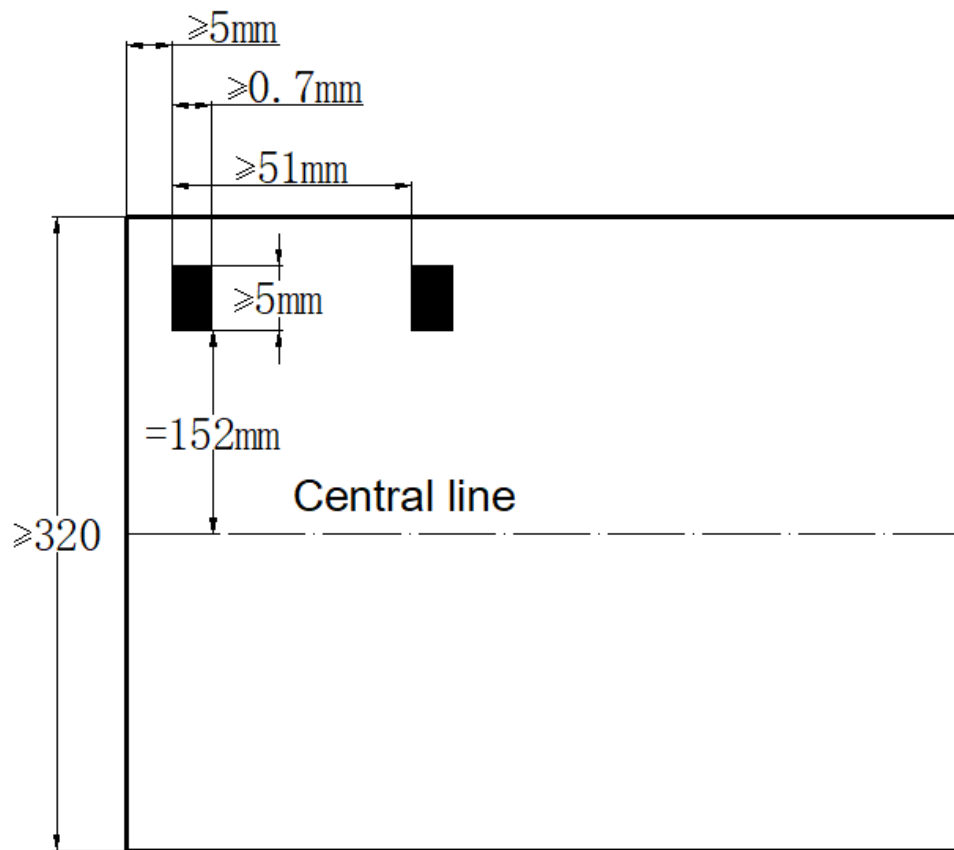


When the marks are designed, following tips should be referred.

1. It is the best to choose white color paper as the printing paper. The detector may not be able to read the sniper mark on another colors of printing paper.
2. The paper size (weight/ thickness) is 157g-350g/0.15-0.35mm, except special papers.
3. It can be used on laminated paper.
4. Crimp degree of paper should less than 2mm. Badly curling paper can lead to precision loss or detecting failure.
5. The optimal number of sniper marks in one paper if from 1 to 10. The tolerance is ± 0.3 mm.
6. Sniper typesetting can be done on the width from 320mm to 330mm or 12.6inch to 13inch of papers.
7. Sniper typesetting can be done on the templates with lead trim, cutter gutter and tail trim.
8. Register mark and sniper mark cannot be used at the same time.
9. Sniper mark requirement (Length*width). The length needs to be greater than or equal to 5mm and the width needs to be greater than or equal to 0.7mm.
10. The color of sniper mark needs to be full black, or it may effect the detecting result. Please take the following color saturation for reference. The recommending value is from 3 to 5.



11. The location of sniper mark and cutting line setting. The width of sniper mark is 0.7mm and the cutting line is on left edge of sniper mark. The cutting tolerance is ± 0.3 mm. The cut compensation can be also made by screen SNIPER FINE 5.2.7
12. The minimum distance between two sniper marks is 51mm.
13. The sniper function can be only used when the paper is wider than or the same to 320mm, and mark position should be as per following drawing



Note: The central line position is according to the central position is found as per 5.6.4. if the infeed is not central, the mark may not work.

5.6 Hardware Setting

5.6.1 Side Guide Setting

Release the two screws circled in red as per following picture. And move the side guide as per the arrow direction.



Place 3-5 sheets on the table and move back side guide to keep the paper, making sure the sheets could move freely in feeding direction but not move in the side guide moving direction.



5.6.2 Skew Adjustment

If side guide position is ok, the feeding is skewed. Release the two screws circled in red as per following picture making sure the screws can move freely in the hole.



Move the screw as per arrow direction in following picture. Move as per the ruler little by little to get the right feeding angle.



5.6.3 Sitters Setting

By loosen the set screws the 4 Slitter head with certain No. can be easily moved along the rod, When it moves to the template mark on the indication plate, The slitter will be stopped by a pot on the rod, softly but appreciably.



331SC feeding guide is not self-central , in order to find the central position, the ruler should be referred



Put a SRA3 paper on the table, when the right side coincide the line of SRA3 of the ruler, the paper is in central position, then set the left side guide.

If the paper is other width, do the same and let the right side coincide the line indicating the width.

If the print has image shift in cross direction, it can be compensated by amend the position according to the ruler.

5.7 Only Trim Sheet Sides

5.7.1 Place Slitters

When only trimming sheet sides, slit #2 and slit #3 will be used for trimming. But they have a limit for trim width **3~6 or over 15 mm**.

If side trim is 7~14 mm or less than 3mm paper piece has some risks going into slitters leading slit jammed. If side trim is over 15 mm paper piece will not go down to the waste box just going out with the finished piece.

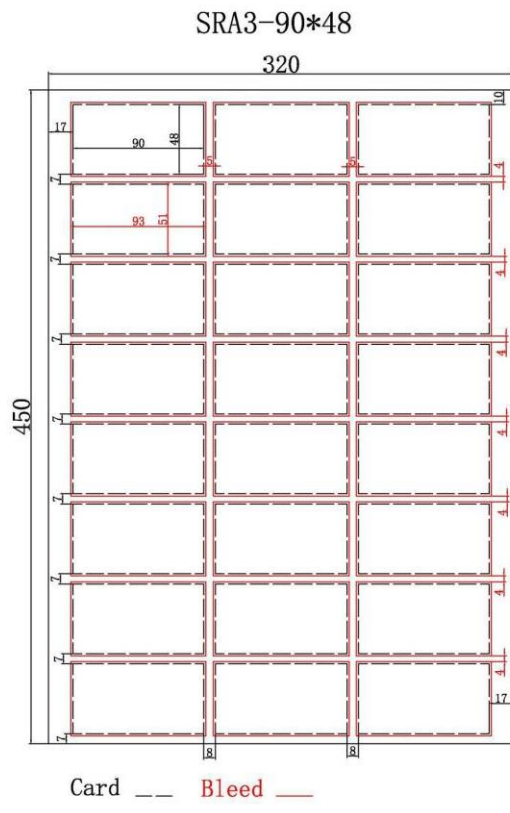
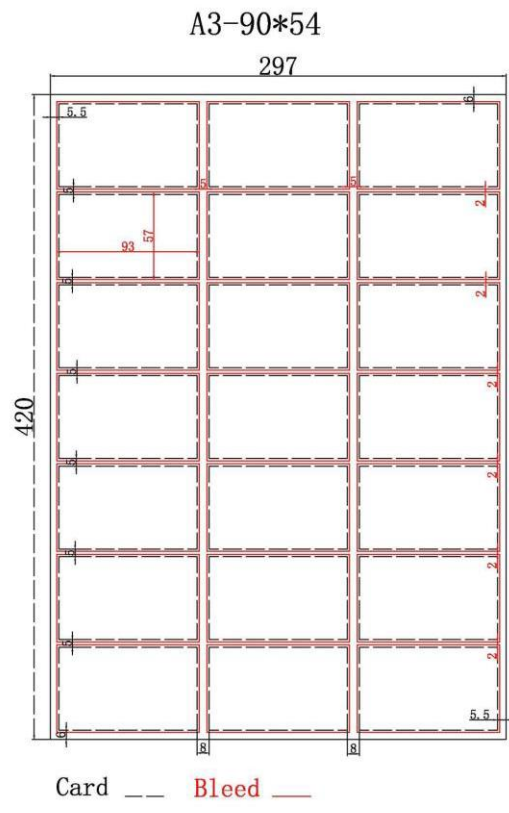
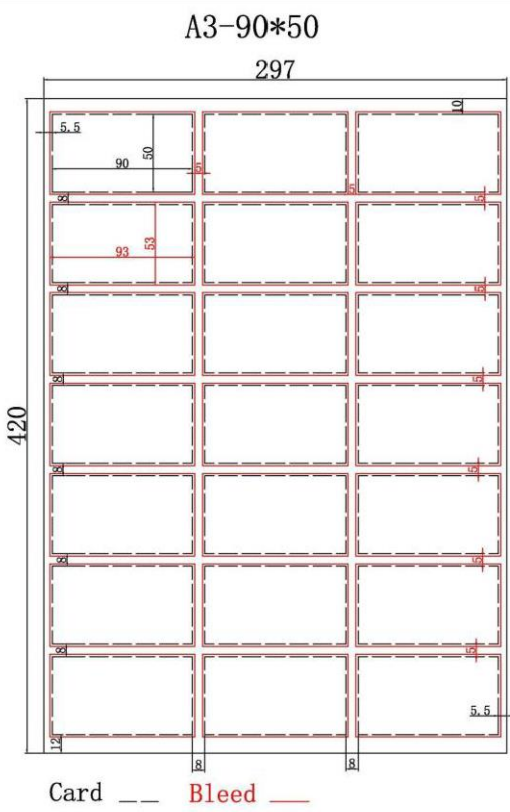
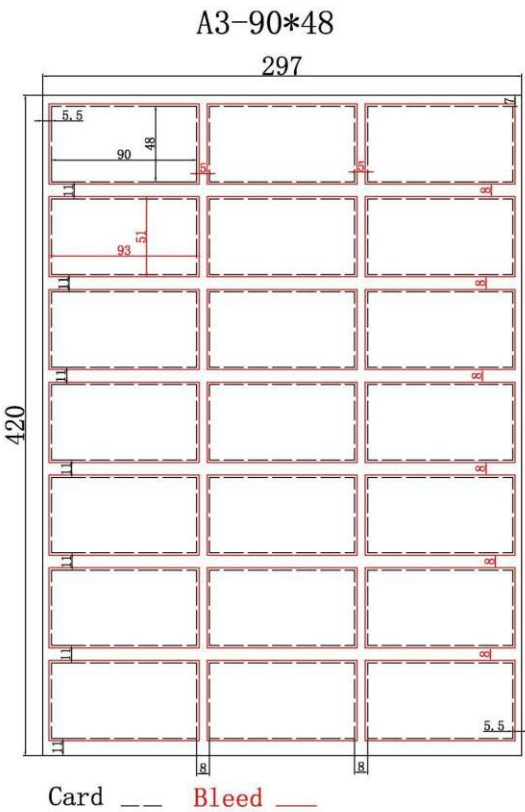
5.7.2 Set Cut Data

Turn off REG. Mark Function if they were open already. Input cut data referred to **5.2.3.5**.

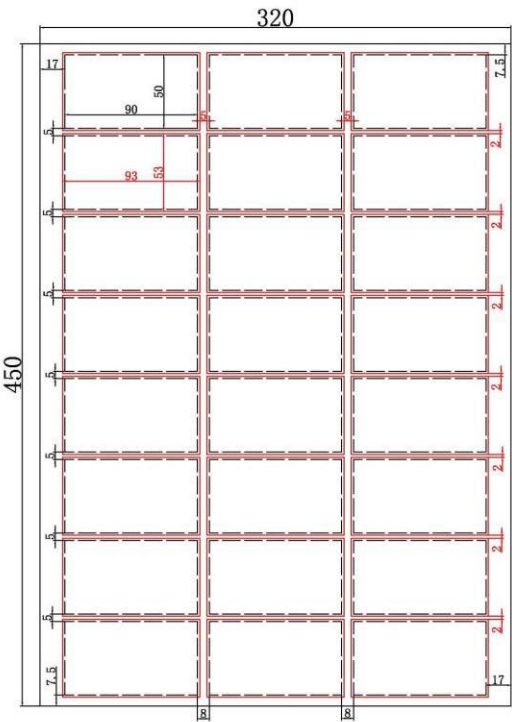
Place the tray properly. Then press Start Key to run machine.

Chapter 6 FACTORY TEMPLATE

6.1 Metric Version

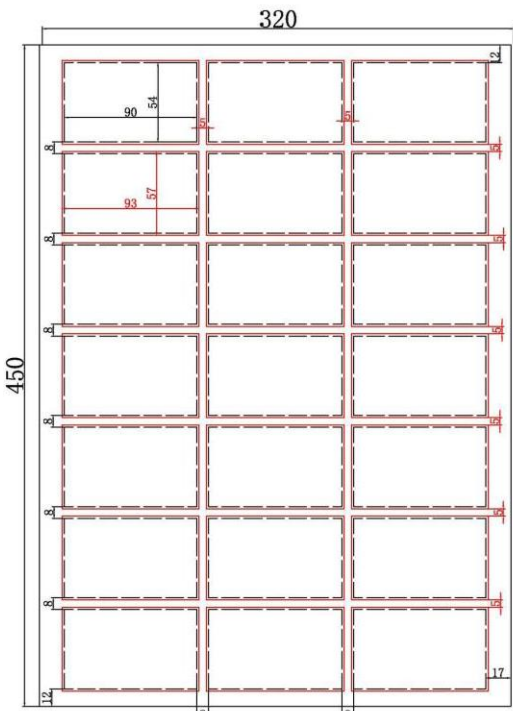


SRA3-90*50



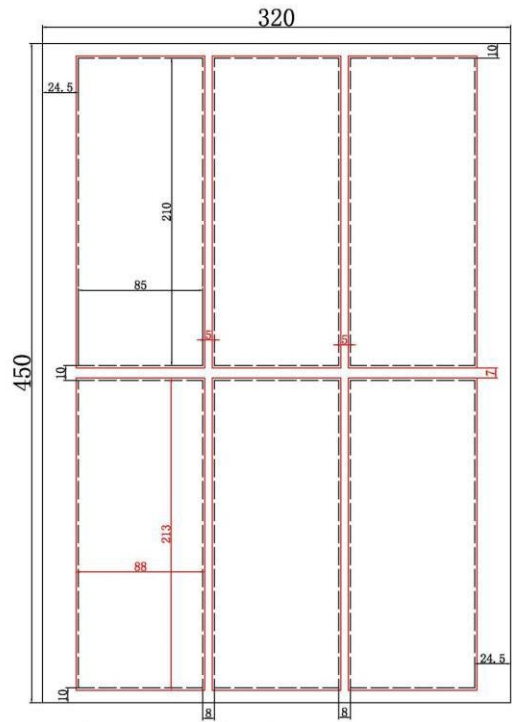
Card __ Bleed __

SRA3-90*54



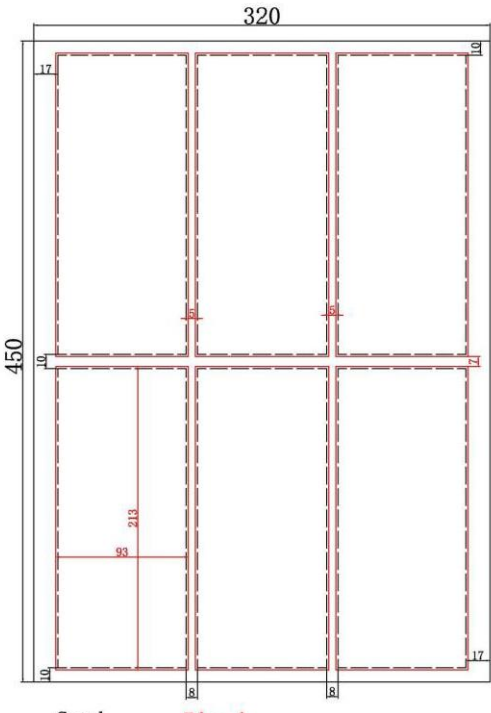
Card __ Bleed __

SRA3-85*210



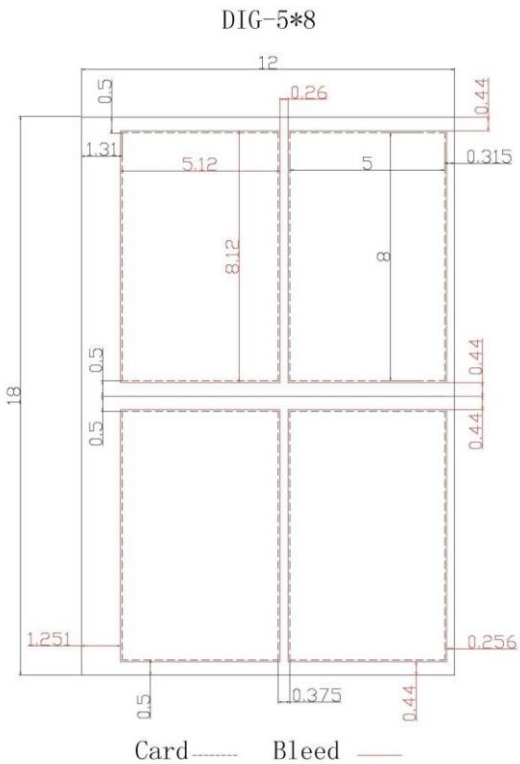
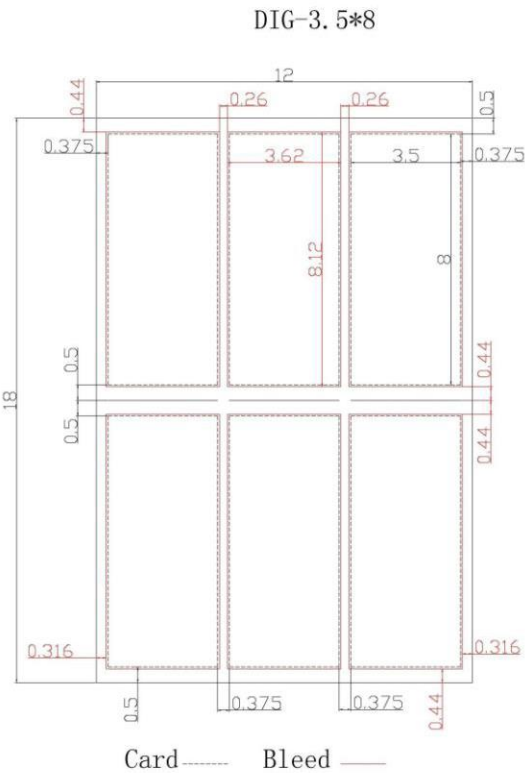
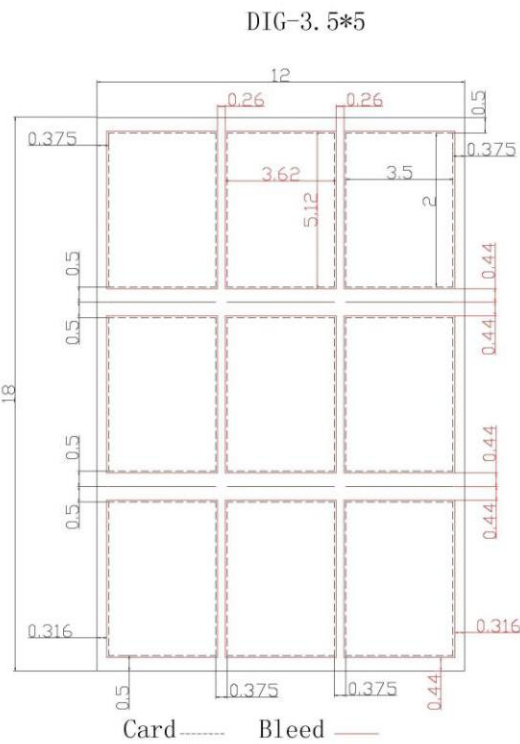
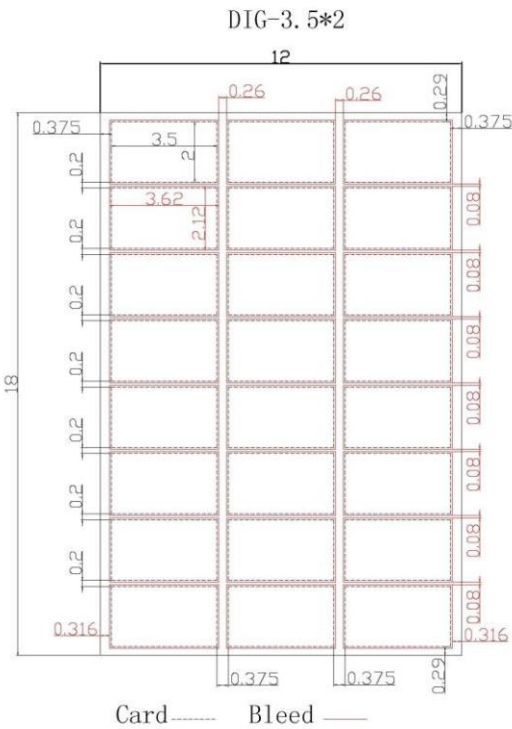
Card __ Bleed __

SRA3-90*210



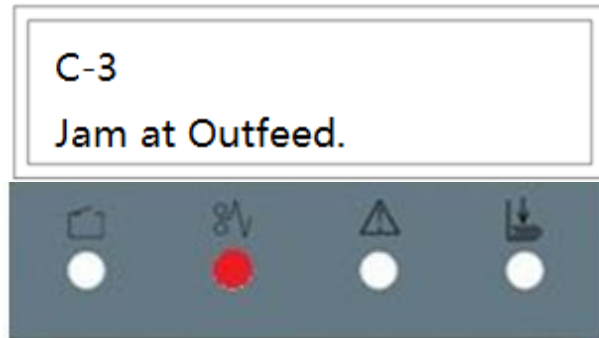
Card __ Bleed __

6.2 Imperial Version

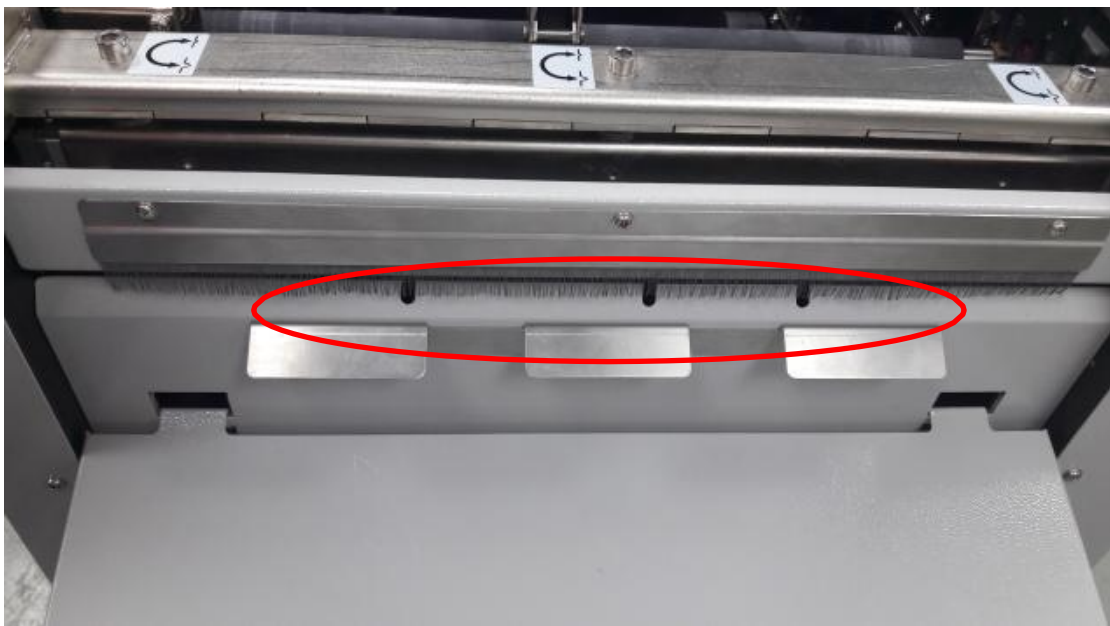


Chapter 7 Trouble Shooting

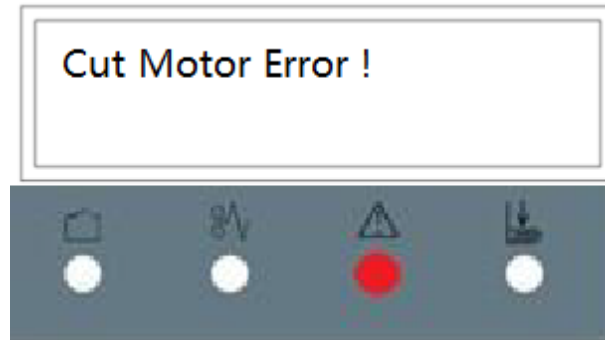
7.1 C-3 Paper Jam at Outfeed Roller



1. Press  or  to drive the jammed paper out.
2. Check if the paper is out of the spec. (100gsm~350gsm).
3. Remove the gauge to check if there is paper piece covering the sensor.
4. Clean the paper dust by air gun on the sensor.



7.2 E-2 Cut Motor Error.



1. Restart the machine.
2. Press or to drive the jammed paper out.
3. Check if the paper is out of the spec. (100gsm~350gsm). As for crease die, the spec. is 100~300gsm.

7.3 Card Cutting non-precise

Result	Reason	Solution
←←←←←← Feeding direction		
 Every card is perfect	 Every cut line is exactly as designed	Perfect No need solution
 Every card is over cut at tail edge	 Image is ahead of template	Compensation by Register mark or Sniper
 Every card is over cut at lead edge	 Image is backward of template	Compensation by Register mark or Sniper
 Latter cut has wider overcut at the tail edge than ones ahead.	 Image is shrinked	Refer 5.2.10 Press to make ratio smaller
 Latter card has wider overcut at the lead edge than ones ahead.	 Image is stretched	Refer 5.2.10 Press to make ratio bigger